Phishing detection
based on machine learning for websites

***Abstract*—Phishing websites pose a significant cybersecurity threat by attempting to deceive users into revealing sensitive and personal information such as financial data or login credentials. This project investigates the application of supervised machine learning techniques for phishing websites detection, using the publicly available dataset "Phishing Website Detector", containing over 11,000 samples and greater than 30 features related to the structure of the URL, webpage behaviour, and domain characteristics. An exploratory data analysis (EDA) was conducted to examine feature relationships and class distribution. Multiple machine learning models such as Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) were evaluated using accuracy, precision, recall and F1-score. Among the models, the Random Forest model achieved the best overall performance, with an accuracy of approximately 96.9%. Further optimization was performed through hyperparameter tuning and feature selection. While tuning resulted in minimal performance change, recall was improved, highlighting the importance of minimizing false negatives within cybersecurity applications.**

Keywords: Phishing, Machine learning, Random Forest, SVM, Logistic Regression, Decision Tree, F1-score

I. INTRODUCTION

Phishing attacks have been a major issue worldwide. The attackers basically create fake websites that can look legit to steal sensitive information. The information could include passwords, login credentials, and personal details. As the world is evolving, we can see that this attack is getting advanced as well, and a lot of people are facing issues with it. There are tools to catch the phishing attack, but nowadays they use different techniques to do the phishing, so it could be hard to detect it, and make it challenging to catch. As you can see, machine learning can help with that because it can see from the data the website contains, and it could tell if the website is real or not. On this project, we made a model, and compared it to other machine learning models to see if it was working properly and if they could tell if a website is fake or not. We also look at false negatives to see which websites are dangerous to go on.

II. BACKGROUND

A. Logistic Regression

Logistic Regression is a statistical technique employed in binary classification tasks [6]. It is a classification technique that uses a logistic (sigmoid) function to predict the probability of an event taking place. This model estimates the sum of all independent variables and maps it to a range of 0 to 1, indicating the likelihood of being in the positive class. The decision boundary can be established using the formula:

$$P(y=1|X) = \frac{1}{1+e^{-(\beta_0+\beta_1 X_1+\ldots+\beta_n X_n)}}$$

Where $P(y=1)X$ is the probability of the data point being in class 1, β_0 is the bias term, and $\beta_1…\beta_n$ are the coefficients for each feature $X_1…X_n$. Logistic Regression is fast and easy between the attributes of phishing detection datasets.

B. Decision Trees

Decision Tree is one of the non-parametric machine learning models used to construct a tree structure suitable for classification problems [7]. It creates a series of partitioning rules to divide the feature space into small parts by choosing the feature with the highest purity (gini impurity, information gain, etc.) recursively until some stopping criteria are met. For example, the Gini impurity for a particular node can be calculated using the following formula:

$$Gini = 1 - \sum_{i=1}^{C} p_i^2$$

Where C is the total number of classes and p_i is the percentage of samples belonging to class i. The lower the Gini impurity, the better. Although Decision Tree models are easy to interpret, they have the risk of overfitting when the depth is not limited.

C. Random Forest

Random Forest is one of the ensemble learning algorithms that utilize a multitude of decision trees to enhance the performance of classification tasks and avoid overfitting [8]. Random Forest constructs a set of decision trees wherein each tree is trained using a bootstrapped version (random sample with replacement) of the input dataset.Furthermore, only a random selection of features is used for each tree split. In the case of classification tasks, the output is computed through the majority vote among all decision trees:

$$\hat{y} = \text{mode}\{T_1(x), T_2(x), \ldots, T_k(x)\}$$

Where $T_i(x)$ represents the output of the i-th tree while k is the total number of trees used in the random forest. It must be noted that Random Forest is a highly suitable machine learning technique for dealing with structured tabular data, especially due to its effectiveness in high dimensional spaces.

D. Support Vector Machine (SVM)

Support Vector Machines (SVM) is a robust supervised learning method used to find the best hyperplane in high dimensional space [9]. The aim of SVM is to maximize the margin, which is essentially the distance between the hyperplane and support vectors - the data points from each class closest to the hyperplane. The decision function is expressed by the equation:

$$\mathbf{w} \cdot \mathbf{x} + b = 0$$

Where $\mathbf{w}$ is the weight vector, $\mathbf{x}$ is the input vector, while b is the bias term. The optimization problem for the linearly separable binary classification case is given by the equation:

$$\min_{\mathbf{w},b} \frac{1}{2}\|\mathbf{w}\|^2 \quad \text{s.t.} \quad y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1, \quad \forall i$$

However, for non-linearly separable data, kernel functions are used by SVM algorithms to map the dataset into higher dimensional space in order to create a linear decision boundary. Despite the fact that SVMs work well for high dimensional data, they may prove to be inefficient in terms of parameter tuning and computational complexity.

E. Evaluation Metrics

Several methods are used to assess the quality of model prediction in order to get a comprehensive idea about its performance. Confusion matrix for binary classification includes TP (phishing detected correctly), TN (Legitimate detected correctly, FP (legitimate detected as phishing), FN (phishing detected as legitimate).

Accuracy is defined by the following formula:

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

However, despite being simple and understandable, it should be noted that accuracy can give an incorrect result on unbalanced data.

Precision is the rate of correct decisions out of positive predictions:

$$Precision = \frac{TP}{TP+FP}$$

It is high if the model rarely produces any false alarm.

Recall is the rate of true positive decisions out of all actual positive instances. Also called sensitivity or true positive rate.

$$Recall = \frac{TP}{TP+FN}$$

As mentioned above, recall is extremely crucial in cybersecurity since the consequences of missing a phishing website are severe for a user.

F1-score is the harmonic mean of precision and recall and is helpful to evaluate both of these characteristics at once:

$$F1\text{-}Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

The F1-score can be used in instances where both metrics are essential and need to be compared.

III. RELATED WORK

Several studies have explored the use of machine learning techniques for phishing website detection using structured datasets and classification models. In [1] the authors proposed a high accuracy phishing detection system using multiple machine learning algorithms, these included Logistic Regression, K-Nearest Neighbors (KNN), Naive Bayes, Random Forest, Support Vector Machine (SVM), and Extreme Gradient Boosting (XGBoost). Their findings show that the Random Forest model is just barely short in performance, putting it at second best, from the XGBoost model that they had tested.

In [2], the authors also used a dataset from Kaggle, it was used to train machine learning models for malicious website detection. Their study addressed class imbalance using sampling techniques which proved the importance of proper data preprocessing in improving model performance. Their approach is similar to ours as we both used publicly available datasets and applied multiple machine learning models to a binary classification problem.

A comparative study done by the authors of [3] evaluated Random Forest, SVM, and Logistic Regression using the accuracy, precision, recall, and F1-score performance metrics. Their finding is similar to our results as theirs also showed that the Random Forest model achieved the best overall performance among their models across the evaluated performance metrics.

Lastly, [4] evaluated KNN, SVM, and Random Forest machine learning models for a phishing detection Google chrome web extension. Their results once again have shown that the Random Forest model provides the best performance for detecting phishing.

Our approach to phishing detection used machine learning models that include Logistic Regression, Decision Tree, Random Forest, and SVM to classify websites as phishing or legitimate using the dataset from Kaggle [5]. While others primarily focused their work on maximizing the overall accuracy, we chose to focus on reducing the false negatives and false positives as they are more important in real world cybersecurity applications.

IV. METHODOLOGY

A. *Dataset Description*

The dataset used in this study was obtained from Kaggle and consists of approximately 11,000 website samples, each described by greater than 30 features. These features capture characteristics of the website such as, but not limited to: URL structure, domain information, security indicators (HTTPS) and webpage behaviour. Each sample is labelled as legitimate (1) or phishing (-1), forming a binary classification.

This dataset provides a suitable representation of both legitimate and illegitimate websites, making it appropriate for training and evaluating machine learning models for phishing detection.The class distribution is relatively balanced, with 6,157 links as legitimate and 4,897 links as phishing, which helps prevent bias towards one class during model training.

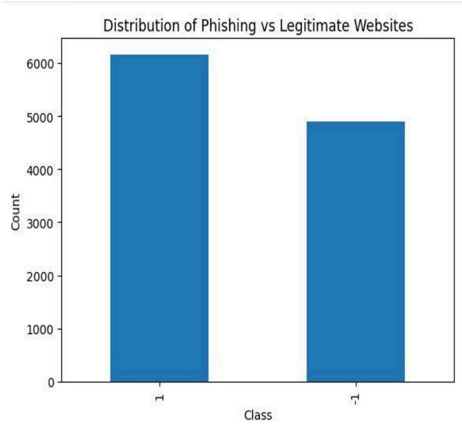

Figure 1. Class distribution chart

B. *Data-preprocessing*

Prior to model training, the dataset was preprocessed to ensure stability and performance. The “Index” column was removed, as it does not contribute to predictive performance. All of the remaining features were numerically encoded, thus no additional encoding or scaling was needed. The dataset was divided into a training and testing subset, using an 80:20 split. The training set was used to train the models, whereas the testing set was reserved for performance evaluation on unseen data.

C. *Machine Learning Models*

In order to evaluate different classification approaches, four multiple machine learning models were implemented and compared: Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM). The models selected represent a wide range of learning strategies including linear, tree based, and kernel based methods.

Logistic Regression was used as a baseline model due to its simplicity and efficiency. It models the probability of a binary outcome using a linear decision boundary [6]. Although it is effective for linear data, its performance is hindered when dealing with complex relationships between features.

TABLE I. RESULTS OF USING LOGISTIC REGRESSION (LR)

Model	Accuracy	Precision	Recall	F1-Score
LR	93.35%	92.97%	95.30%	94.12%

The Decision Tree classifier models classification through a hierarchical structure of decision rules based on feature splits [7]. It provides a clear decision path, making it highly interpretable. The problem with decision trees is that they are prone to overfitting data, especially when the tree depth is not constrained, which reduces performance.

TABLE II. RESULTS OF USING DECISION TREE (DT)

Model	Accuracy	Precision	Recall	F1-Score
DT	95.84%	96.43%	96.11%	96.27%

Random Forest is a machine learning method that combines multiple decision trees in order to improve overall performance. Each tree is trained on a random subset of data and features, and the final prediction is determined through voting [8]. The Random Forest approach reduces the overfitting problem that Decision Tree has, and improves the accuracy. Because of this advantage, Random Forest is well suited for structured tabular datasets like the "Phishing Website Detector”.

TABLE III. RESULTS OF USING RANDOM FOREST (RF)

Model	Accuracy	Precision	Recall	F1-Score
RF	96.97%	96.79%	97.81%	97.30%

Support Vector Machine (SVM) is a kernel based classification method that finds an optimal decision boundary separating the classes [9]. SVM is effective in handling high dimensional data and can model complex decision boundaries, however, SVM can be extremely sensitive to parameter selection, requiring additional tuning for the optimal performance.

TABLE IV. RESULTS OF USING SVM

Model	Accuracy	Precision	Recall	F1-Score
SVM	95.11%	94.13%	97.33%	95.70%

D. Model Evaluation Metrics

As observed from the previous tables, multiple metrics were used in order to evaluate the performance of each of the models. These metrics include: accuracy, precision, recall and F1-score.

Accuracy measures the overall proportion of the correctly classified samples. Precision measures how accurately the model identifies phishing websites when predicting a positive class. Recall measures the model's ability to correctly identify all phishing websites in a given set, making it the most important in cybersecurity applications where false negatives have serious consequences. F1-score provides a balanced measurement by combining both precision and recall together.

In addition to those metrics, confusion matrices were used to provide a detailed description of performance, highlighting the true positives, true negatives, false positives, and false negatives. The performance matrix of the best performing model is shown below, showing only 40 false negatives, and 27 false positives.

E. Hyperparameter Tuning

In order to further improve the model performance of Random Forest, hyperparameter tuning was applied. GridSearchCV was used to systematically explore different combinations of hyperparameters, including the number of trees used, tree depth, and minimal samples required to split and leaf nodes.

Two tuning strategies were implemented, using different scoring metrics. Initially, the F1-score was used as the objective, in order to balance precision and recall. Following that optimization, recall was used as the scoring metric to prioritize the detection of phishing websites. This approach reflects the importance of minimizing false negatives within cybersecurity applications.

The tuned model was then evaluated on the test dataset, and compared to the baseline RF model to assess the impacts of its optimization.

F. Feature Selection

Feature selection was performed to identify the most important features contribution to performance. The feature importance scores were obtained from the TF model, and the top features were selected based on their contribution.

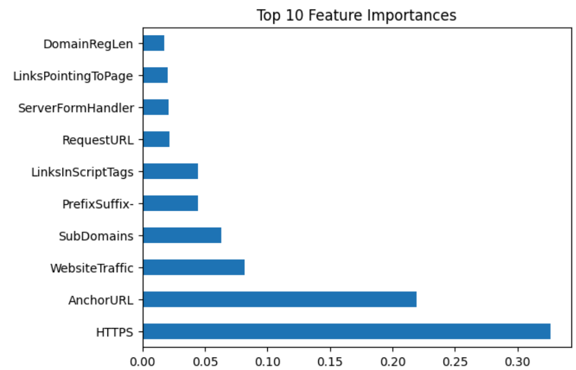

Figure 2. Feature importance graph

As shown above, the graph reveals that some features such as HTTPS status, Anchor URL, and website traffic contributed the most to performance. This is accurate as security indicators and suspicious links are commonly associated with fraudulent (phishing) websites.

A reduced dataset was created using the most important features, and the RF model was retrained using this subset. The aim of this approach was to reduce redundancy, simplify the model, and improve the overall computational efficiency.

V. RESULTS

A. Model Performance Comparison

To evaluate the effectiveness of different machine learning approaches, the four classification models were implemented and tested. The performance of each model was evaluated using accuracy, precision, recall and F1-score. The following table is a summary of the performance of all models.

TABLE V. SUMMARY COMPARISON OF ALL MODELS

Model	Accuracy	Precision	Recall	F1-Score
LR	93.35%	92.97%	95.30%	94.12%
DT	95.84%	96.43%	96.11%	96.27%
RF	**96.97%**	**96.79%**	**97.81%**	**97.30%**
SVM	95.11%	94.13%	97.33%	95.70%

From the results, it is clear that Random Forest achieved the highest accuracy and F1-Score, indicating its superior performance to the other methods. Logistic Regression showed the lowest performance, which can be attributed to its linear nature and inability to capture complex relationships within a dataset. Decision Tree performed well, although it is prone to overfitting compared to the other methods. SVM demonstrated a strong recall, but slightly lower performance compared to Random Forest.

B. Confusion Matrix Analysis

To further evaluate the performance of the best model, a confusion matrix was analyzed for the Random Forest model.

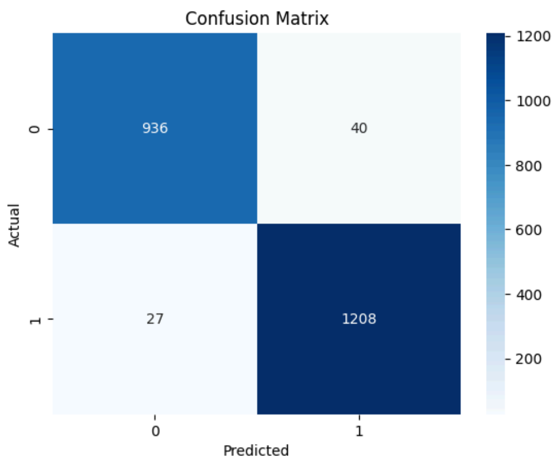

Figure 3. Confusion matrix of the RF model

The confusion matrix shows that the majority of phishing and legitimate websites were correctly classified, with only 40 false negatives, and 27 false positives. The model achieved 936 true positives (correctly identified phishing) and 1208 true negatives (correctly identified legitimate).

False negatives, which represent phishing websites misclassified as legitimate, were minimal. This is the most important observation, as false negatives pose a significant risk in cybersecurity applications as they allow malicious websites to go undetected. Having a low number of false positives also proves that the model does not excessively misclassify legitimate websites as phishing.

C. Hyperparameter Tuning Results

To further improve performance, hyperparameter tuning was applied to the RF model, using GridSearchCV. Different configurations of model parameters were explored, including the number of trees, maximum tree depth, and minimal samples required for splitting.

Tuning was performed using the F1-score as the optimization, then later changed to recall, in order to prioritize the detection of phishing websites.

TABLE VI. RESULTS OF USING RF vs TUNED RF

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Base	96.97%	96.79%	97.81%	97.30%
Tuned	96.87%	96.56%	97.89%	97.22%

The tuning model achieved a slight improvement in recall, indicating improved detection of phishing websites. However with this improvement, came with a minor decrease in accuracy as well as precision. In the context of phishing detection, this trade off is acceptable, as minimizing false negatives is more critical than achieving minimal gains in accuracy.

D. Feature Selection Analysis

Feature selection was applied using feature importance scores derived from the RF model. The top features were selected in order to reduce redundancy and improve the models efficiency.

The model was retrained using only selected features, and its performance was evaluated with the baseline, as well as tuned models.

TABLE VII. RESULTS OF USING RF vs TUNED RF vs FEATURE SELECTED (FS)

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Base	96.97%	96.79%	97.81%	97.30%
Tuned	96.87%	96.56%	97.89%	97.22%
FS	96.06%	95.63%	97.41%	96.51%

The feature selected model showed an overall slight decrease in performance compared to the other two models. This suggests that although some features were less important, they still contributed useful information in the classification process.

E. Summary of Findings

Overall, Random Forest was identified to be the most effective model for phishing website detection within this study. It consistently outperformed the other models across all evaluation metrics, and demonstrated strong generalization performance. Hyperparameter tuning resulted in marginal improvements, specifically in recall, which puts emphasis on the importance of optimizing models based on application specific priorities. Feature selection was implemented, but introduced a performance trade off. These results demonstrate that machine learning methods, specifically Random Forest, are highly suitable for phishing detection datasets.

VI. CONCLUSION

This project shows how we use machine learning to deal with phishing attacks. We used a dataset that can help detect phishing, which used 11,000 website samples and uses 30 features that include URL structure, domain information, security indicators (HTTPS), and webpage behaviour. The models we use are Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine (SVM) to compare the models in terms of Accuracy, Precision, Recall, and F1-score. The model we pick is Random Forest, which has the highest result, and this can help us detect phishing websites. We also changed the settings of the model because some phishing websites have advanced features and settings to see if it was working properly. We also check different types of features for our model, so we can see which one best fits our model. Overall, as you can see, for machine learning, we should pick Random Forest as the best choice for detecting phishing websites, and also, this model can be more advanced when we change its structure to detect challenging websites in the future.

VII. APPENDIX

A. Code Repository

https://github.com/wheresak/MachineLearningClass

B. Group Contribution

Adam Khan: Coding, Abstract, Methodology, Results
Shariq Haider: Introduction, Conclusion
Abdul Hannan: Background
Rayyan Malik: Related Work

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